

9(a)	any three from: <i>less hexokinase so:</i> 1 less glucose, phosphorylated / activated ; 2 less glucose diffuses into cell (as less steep concentration gradient) ; 3 less, glycolysis / formation of pyruvate ; 4 less ATP formed ; 5 loss of red blood cells / anaemia / less oxygen transport ; <i>plus</i> <i>symptom</i> 6 fatigue / shortness of breath / faint / feeling dizzy ;	4
9(b)	any four from: 1 acetyl CoA / acetyl group / acetate / 2C fragment, combines with oxaloacetate to form citrate ; 2 decarboxylation / carbon dioxide produced ; 3 dehydrogenation / oxidation (of named intermediate) ; 4 reduced, NAD / FAD, produced ; 5 ATP produced by substrate-linked <u>phosphorylation</u> ; 6 oxaloacetate, recycled / regenerated / formed again ;	4

Question	Answer	Marks
9(c)	any three from: 1. pyruvate / (net 2) ATP, formed by glycolysis or substrate-linked phosphorylation occurs in glycolysis ; 2 pyruvate reduced by reduced NAD ; Reject pyruvate is dehydrogenated 3 lactate formed / lactate fermentation ; A lactic acid 4 NAD, regenerated / recycled or reduced NAD oxidised (to NAD) / reduced NAD converted to NAD ; 5 AVP ; e.g. lactate dehydrogenase	3